Answer Chapter 1 — Web Development

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A. Multiple Choice Questions

1. (d) Web is an information service running on the Internet.

2. (a) Sending email from a desktop client.

3. (b) Browsers must implement shared open standards.

4. (c) Translates domain names into IP addresses.

5. (c) A photographer's portfolio.

6. (d) It lets users perform tasks on their own data.

7. (b) Implementing authentication and APIs.

8. (c) Client-side code can be bypassed by the user.

9. (c) Requirements, design, planning, testing...

10. (d) Wireframe.

11. (d) Working features delivered in short iterations.

12. (a) Correct capitalization of the requested filename.

B. True/False Questions

1. False — The Internet existed before the Web.

2. True

3. False — Static sites can still run client-side JavaScript.

4. False — Browser checks can be bypassed, so server validation is needed.

5. True

6. False — Waterfall moves through phases in sequence.

7. False — DevTools edits are temporary and local.

8. True

C. Short Answer Questions

1. Internet vs. Web: The Internet connects networks worldwide. The Web is a service that runs on it using HTTP and HTML.

2. Page Load Sequence: Enter URL → DNS lookup → TCP/TLS connection → HTTP request → server response → render HTML, CSS, and JavaScript.

3. Static vs. Dynamic Websites: Static sites serve fixed files. Dynamic sites generate content using server logic and often a database.

4. Waterfall vs. Agile: Waterfall uses sequential phases. Agile delivers features in short cycles and uses feedback.

5. Developer Tools Panels: Elements inspects the page, Console shows logs/errors, Network tracks requests, and Device Mode checks responsive layouts.

D. Essay Questions

1. URL Execution Lifecycle: The browser checks DNS, connects to the server, sends an HTTP request, receives a response, and displays the page and its resources.

2. Hospital Website Requirements: Static pages show the address and working hours. Dynamic pages show doctor schedules. The patient portal handles appointments and records with secure login.

3. Six-Month Redesign Plan: Months 1–2: planning and wireframes. Months 3–4: build responsive pages. Month 5: connect the portal and database. Month 6: test, fix security issues, and launch.

E. Practical Programming Questions

1. Requirements — Seminar Page: Functional: show seminar details, validate registration, display a countdown, and link to a PDF schedule. **Non-functional:** load quickly and work on mobile screens.

2. Network Audit: Page A: 200 OK, 14 requests, 420 KB, 0.8 s. **Page B:** 200 OK, 48 requests, 3.8 MB, 4.2 s. Page A is better for slow connections because it loads faster and uses fewer resources.

3. Sitemap: Home (index.html), About (about.html), and Join (join.html). The Join page contains the member application form.

4. Mobile Responsive Audit: At 360px, the navigation overflows and the image causes horizontal scrolling. Use `max-width: 100%; height: auto;` and allow navigation items to wrap.

5. Test Plan — 8 Cases: 1) Submit valid form: success response. 2) Empty form: validation message. 3) Press Tab: focus visible. 4) Resize to 360px: single column. 5) Check images: alt text exists. 6) Reload: cached assets may return 304. 7) Click footer link: no 404. 8) Profile load: scripting under 50 ms.

6. Diagnostic & Redesign Case Study: The mobile site fails because it uses a fixed 960px width. Redesign it with a collapsible header, a responsive hero section, meeting details, and a full-width registration button.